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Strap adjustment system for a cosmetic applicator configured for users with limited mobility

Abstract

A system that facilitates adjustment of a strap attached to a cosmetic applicator configured for users with limited mobility. The system allows adjustment via a single hand if needed, and can be used by users missing one or more digits. The strap adjustment system includes a housing cap, a strap, a locking feature, and a roller wheel, the locking feature disposed on a first side of a chamber formed by the housing cap and an end of the cosmetic applicator, the roller wheel disposed on a second side of the chamber, and the strap configured to pass through the chamber with the locking feature configured to contact a first side of the strap and lock the strap against locking edges of the chamber. The roller wheel is configured to unlock the strap and adjust the strap via rotation of the roller wheel.

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Background/Summary

FIELD

(1) The present disclosure describes a system and features related to a device for modifying, mitigating, altering, reducing, compensation for, or the like, the movement of a cosmetic applicator caused by unintentional movements, tremors, limited mobility, or the like of a user.

BACKGROUND

(2) Unintentional movements of the human body, or human tremors, can occur in individuals suffering from motion disorders or even healthy individuals. Due to these unintentional movements, a person may have difficulty in performing a task that requires care and precision, such as applying a cosmetic composition to a part of the body, such as the face, hands, or feet.

(3) Therefore, there is a need for a solution that allows application of a cosmetic composition that is compatible with the diverse and disposable nature of cosmetic applicators.

SUMMARY

(4) In one embodiment, the present disclosure relates to a strap adjustment system for a cosmetic applicator configured for users with limited mobility, including a housing cap disposed at an end of

the cosmetic applicator, the housing cap and the end of the cosmetic applicator forming a chamber including a first opening and a second opening; a strap attached to the cosmetic applicator at a first end of the strap, a second end of the strap configured to pass through the first opening and the second opening; a locking feature disposed on a first side of the chamber and configured to (i) contact a first side of the strap, (ii) bias the strap in a first direction away from the first side of the chamber via a locking force, and (iii) force a second side of the strap to abut a first locking edge and a second locking edge of the chamber formed, the first locking edge formed as part of the first opening of the chamber and second locking edge formed as part of the second opening of the chamber, the locking force preventing translation of the strap through the chamber; and a roller wheel disposed on a second side of the chamber opposite the first side of the chamber, the roller wheel configured to (i) abut the second side of the strap when an unlocking force is applied in a second direction opposite the first direction to force the roller wheel into an engaged state, the unlocking force being sufficient to overcome the locking force applied by the locking feature and allow the strap to translate through the chamber, (ii) rotate in the engaged state to translate the strap through the chamber, and (iii) return to an unengaged state where the roller wheel does not provide sufficient unlocking force to overcome the locking force applied by the locking feature on the strap.
(5) In an embodiment, the locking feature is a reversibly deformable structure configured to apply the locking force in the first direction via a spring force.
(6) In an embodiment, the locking feature is a magnet configured to apply the locking force in the first direction via a magnetic force.
(7) In an embodiment, the strap is reversibly deformed in the chamber via the locking force applied by the locking feature, the deformation of the strap increasing a friction force between the strap at the first locking edge and the second locking edge.
(8) In an embodiment, the roller wheel is reversibly biased away from the strap in the first direction.
(9) In an embodiment, the roller wheel includes an axle disposed through a rotation axis of the roller wheel, a first end of the axle and a second end of the axle being disposed in a first axle housing and a second axle housing, respectively, formed as part of the second side of the chamber, the first axle housing and the second axle housing each including an elongated opening along the first and second direction through which the first end of the axle and the second end of the axle can travel between the engaged and unengaged states of the roller wheel.
(10) In an embodiment, the roller wheel is arranged on a swiveling platform formed as part of the second side of the chamber, the swiveling platformed configured to rotate about an axis orthogonal to a rotation axis of the roller wheel and rotate the roller wheel along a direction of an applied force applied to the roller wheel.
(11) In an embodiment, the strap includes friction features on the second side of the strap configured to provide additional resistance force against translation of the strap through the chamber.
(12) In an embodiment, the roller wheel includes complementary friction features as the friction features on the second side of the strap, the complementary friction features of the roller wheel being disposed on a surface of the roller wheel and configured to abut and mate with the friction features of the strap.
(13) In an embodiment, the roller wheel is configured to translate the strap to a first resistance checkpoint.
(14) In an embodiment, the first end of the strap is attached to the cosmetic applicator via an adjustable strap attachment configured to reversibly translate in the first direction and the second direction.
(15) In an embodiment, after reaching the first resistance check point, the roller wheel is configured to further translate the strap through the chamber and adjust an arrangement of the adjustable strap attachment along the first and second direction.

(16) In an embodiment, the roller wheel is configured to translate the strap to a second resistance checkpoint after which a tightness of the strap around an object disposed within a loop of the strap cannot be tightened further.
(17) The present disclosure additionally relates to a method of adjusting a strap in a strap adjustment system for a cosmetic applicator configured for users with limited mobility, including applying an unlocking force towards a housing cap disposed at an end of the cosmetic applicator, the housing cap and the end of the cosmetic applicator forming a chamber including a first opening and a second opening, the strap attached to the cosmetic applicator at a first end of the strap, a second end of the strap configured to pass through the first opening and the second opening, the strap adjustment system additionally including a locking feature disposed on a first side of the chamber and configured to (i) contact a first side of the strap, (ii) bias the strap in a first direction away from the first side of the chamber via a locking force, and (iii) force a second side of the strap to abut a first locking edge and a second locking edge of the chamber formed, the first locking edge formed as part of the first opening of the chamber and second locking edge formed as part of the second opening of the chamber, the locking force preventing translation of the strap through the chamber, and a roller wheel disposed on a second side of the chamber opposite the first side of the chamber, the roller wheel configured to (i) abut the second side of the strap when the unlocking force is applied in a second direction opposite the first direction to force the roller wheel into an engaged state, the unlocking force being sufficient to overcome the locking force applied by the locking feature and allow the strap to translate through the chamber, (ii) rotate in the engaged state to translate the strap through the chamber, and (iii) return to an unengaged state where the roller wheel does not provide sufficient unlocking force to overcome the locking force applied by the locking feature on the strap; and while applying the unlocking force towards the housing cap sufficient to overcome the locking force, applying a lateral force to rotate the roller wheel and translate the strap through the chamber.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) A more complete appreciation of the embodiments and many of the attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings, wherein:
(2) FIG. **1** shows a motion stabilizing device, according to an embodiment of the present disclosure.
(3) FIG. **2** shows how the motion stabilizing device couples with an adaptor and a make-up applicator, according to an embodiment of the present disclosure.
(4) FIG. **3**A shows a diagram of the internal components of a motion stabilizing, according to an embodiment of the present disclosure.
(5) FIG. **3**B shows a diagram of an alternative embodiment of the motion stabilizing device in which a receiver portion includes an electromagnetic positioner, according to an embodiment of the present disclosure.
(6) FIG. **4** shows an overview of a universal adapter handle connection system, according to an embodiment of the present disclosure.
(7) FIG. **5** shows an illustration of a strap adjustment system with friction locks, according to an embodiment of the present disclosure.
(8) FIG. **6** shows an illustration of a strap adjustment system with a locking pin, according to an embodiment of the present disclosure.
(9) FIG. **7** shows an illustration of a strap adjustment system with a ratchet button, according to an

embodiment of the present disclosure.
(10) FIG. **8** shows an illustration of a strap adjustment system with automated strap adjustment, according to an embodiment of the present disclosure.
(11) FIG. **9** shows an illustration of the strap adjustment system with the adjustable or dynamic attachment of the strap, according to an embodiment of the present disclosure.
DETAILED DESCRIPTION
(12) The present disclosure describes a cosmetic applicator system that minimizes, modifies, mitigates, alters, reduces, compensates for, or the like unintentional movements by stabilizing, orienting, operating, controlling, etc. an applicator for a user and is also designed to be flexible to accommodate different types of commercially available cosmetic applications. The present disclosure further describes a system and features to enhance the functionality of such a cosmetic applicator system.
(13) The basic features and operation of a motion stabilizing device for a cosmetic applicator is described in U.S. Pat. No. 11,458,062, which is incorporated herein by reference.
(14) FIG. **1** shows a conventional motion stabilizing device **1100**, which serves as a base unit for receiving a cosmetic applicator according to an embodiment. The device **1100** includes a handle portion **1101**, a receiver portion **1102**, and a strap **1103**. The receiver portion **1102** includes an interface **1104**, shown as a male connector that couples with a cosmetic applicator, which will be discussed in detail below. The receiver portion could be utilized for communication between the base unit and the applicator. The connection to an adaptor and/or an applicator could be accomplished with a mechanical coupling, such as screw-in or snap-fit, or it could be accomplished with magnets.
(15) FIG. **2** shows how the device **1100** couples with an adaptor **1105** and a make-up applicator **1106**. It can be seen that the adaptor fits over the exposed end of the receiver portion **1102**. The adaptor includes electrical mating connectors (a female connector—not shown) in a recessed portion to make contact with the electric interface of the receiver portion **1101**.
(16) As shown in FIG. **2**, the receiver portion **1102** is configured to contort, articulate, reposition, etc., between an upright posture (as shown in FIG. **1**) and an angled posture (as shown in FIG. **2**). This is accomplished with a hinge mechanism contained inside the receiver portion **1102**. FIG. **2** shows that the hinge mechanism is a self-leveling/motion stabilizing hinge.
(17) FIG. **3**A shows a diagram of the internal components of device **1100** according to one embodiment. In the handle portion, the device includes a power source **1301**, which may be a battery or the like. The device includes a printed circuit board assembly (PCBA) **1302**, which may include positional sensor circuitry **1307**, reader circuitry **1308**, control circuitry **1309**, and communication interface **1310**, as understood in the art.
(18) For instance, as the sensor circuitry **1307**, the PCBA may include at least one inertial sensor and at least one distributed motion sensor to detect unintentional muscle movements and measure signals related to these unintentional muscle movements that are created when a user adversely affects motion of the applicator. These sensors also detect the motion of the stabilized output relative to device. The control circuitry sends voltage commands in response to the signals to the motion generating elements (described below) to cancel the user's tremors or unintentional muscle movements. This cancellation maintains and stabilizes a position of the applicator, keeping it stable.
(19) One of ordinary skill in the art readily recognizes that a system and method in accordance with the present invention may utilize various implementations of the control circuitry and the sensor circuitry and that would be within the spirit and scope of the present invention. In one embodiment, the control circuitry **1309** comprises an electrical system capable of producing an electrical response from sensor inputs such as a programmable microcontroller or a field-programmable gate array (FPGA). In one embodiment, the control circuitry comprises an 8-bit ATMEGA8A programmable microcontroller manufactured by Atmel due to its overall low-cost, low-power

consumption and ability to be utilized in high-volume applications.
(20) In one embodiment, the at least one inertial sensor in the sensor circuitry is a sensor including but not limited to an accelerometer, gyroscope, or combination of the two. In one embodiment, the at least one distributed motion sensor in the sensor circuitry is a contactless position sensor including but not limited to a hall-effect magnetic sensor.
(21) The system created by the combination of the sensor circuitry, the control circuitry, and the motion generating elements may be a closed-loop control system that senses motion and acceleration at various points in the system and feeds detailed information into a control algorithm that moves the motion-generating elements appropriately to cancel the net effect of a user's unintentional muscle movements and thus stabilize the position of the applicator. The operation and details of the elements of the control system and control algorithm are understood in the art, as described in U.S. PG Publication 2014/0052275A1, incorporated herein by reference.
(22) The communication interface **1310** may include a network controller such as BCM43342 Wi-Fi, Frequency Modulation, and Bluetooth combo chip from Broadcom, for interfacing with a network.
(23) In the receiver portion of the device, there may be two motive elements to allow 3-dimensional movement of the receiver as anti-shaking movement. The two motive elements include a y-axis motive element **1303** and an x-axis motive element **1304**, each being connected to and controlled by the PCBA **1302**. Each of the motive elements may be servo motors as understood in the art. The device further includes end effector coupling **1305**, which is configured to couple with the adaptor **1105**. The end effector coupling **1305** may include a radiofrequency identification (RFID) reader **1306**, configured to read an RFID tag, which may be included with the applicator, as will be discussed below.
(24) FIG. **3**B shows a diagram of an alternative embodiment of the device **1100** in which the receiver portion includes an electromagnetic positioner 1311 instead of the motive elements shown in FIG. **3**A. The electromagnetic positioner 1311 may include U-shaped magnetic cores **1312** arrayed around a non-magnetic tube **1313**, which is filled with a magnetic fluid **1314**. Each of the magnetic cores has arm portions that are surrounded by windings **1315**. The magnetic cores may be controlled by the control circuitry in the PCBA **1302** to act as a controllable active magnetic field-generating structure which is used to generate a variable magnetic field that acts upon the magnetic fluid, causing it to be displaced, thereby enabling the armature to be moved to a desired coordinate position and/or orientation. The details of implementing the electromagnetic positioner 1311 may be found in U.S. Pat. No. 6,553,161, which is incorporated herein by reference.
(25) In the above-described conventional motion stabilizing device, there is a problem that the interface **1104** that receives the adaptor **1105** requires a specific point of attachment to align properly with the interface.
(26) Therefore, the below embodiments provide a universal adapter connection between the handle of the motion stabilizing device in order to improve user experience and reduce the struggle and time taken to set up the system for use.
(27) In one embodiment, the present disclosure is directed towards a cosmetic applicator. The cosmetic applicator can be used for a variety of cosmetics and cosmetic applications, including, but not limited to, mascara, eyeliner, eyebrow products, lip products (lipstick, lip gloss, lip liner, etc.), skin products, and/or hair products. In one embodiment, the cosmetic applicator can include an adapter, wherein the adapter can connect the cosmetic applicator to a motion stabilizer. The motion stabilizer can be, for example, a handle that can counteract unintentional motions such as tremors or spasms. These motions can interfere with the application of cosmetics and can also make it difficult to generally interact with cosmetic applicators or tools. For example, the many cosmetic products require a twisting motion or force to be applied to open or extrude the product. It can be difficult for users to achieve the range of motion or the precision necessary to apply these forces to the cosmetic. In one embodiment, the cosmetic applicator can hold a cosmetic and can enable the

proper force to be applied to the cosmetic to open, close, mix, stir, blend, extrude, or achieve other similar functions necessary for application.

(28) FIG. **4** shows an overview of a cosmetic securement device including a motion stabilizer device **150** and a cosmetic adapter **100**, according to one embodiment of the present disclosure. The motion stabilizer device **150** can include at least one of a handle portion **151** and a hinge portion **152** (receiver portion). In an embodiment, the handle portion **151** is functionally similar to the handle portion **1101** shown in FIG. **1**. In an embodiment, the receiver portion **152** is functionally similar to the receiver portion **1102** shown in FIG. **1**. The motion stabilizer device **150** as a whole is referred to as a motion stabilizer handle. In an embodiment, the cosmetic securement device includes a cosmetic adapter **100**, wherein the adapter **100** is coupled to the receiver portion **152** of the motion stabilizer device **150**. The cosmetic adapter **100** is configured to hold a cosmetic tool or product. In one embodiment, the cosmetic adapter **100** comprises a universal adapter configured to hold different types of cosmetic products. For example, the adapter **100** can include a ring-shaped holder. Various configurations of the adapter **100** can be used for different cosmetics and tools. For example, the shape and dimensions of the cosmetic holder can be configured for different cosmetic products. The adapter **100** can be easily attached to and removed from the motion stabilizer device. The basic features and operation of the cosmetic securement device of FIG. **4** are described in co-pending U.S. patent application Ser. Nos. 18/091,882; 18/091,920; 18/091,843; 18/091,925; 18/148,957; and Ser. No. 18/148,880; and Ser. No. 18/148,930, which are incorporated herein by reference.

(29) In one embodiment, the adapter **100** is coupled to the receiver portion **152** of the motion stabilizer device **150** via a magnetic attachment. The base or bottom end of the adapter **100** forms a chamber that can fit over a projection at the tip of the receiver portion **152**. The base of the adapter includes at least one magnet, wherein the at least one magnet can be attracted to and attach to a magnet in the receiver portion **152**. In one embodiment, the chamber formed at the base of the adapter **100** can fit over the receiver portion **152** in more than one orientation. The chamber can be a hollow area within the body of the adapter that is fully contained by the walls of the adapter **110**. The chamber can be approximately conical in shape. In one embodiment, the chamber is cylindrical. The chamber can be configured to fit over a projection on one end of the motion stabilizer. The chamber can be used to align and guide the attachment of the cosmetic applicator to the motion stabilizer. The fit of the chamber over and around the projection on the end of the motion stabilizer limits lateral motion that would misalign the cosmetic applicator and the motion stabilizer. It can be easier for a user to align the chamber over the end of the motion stabilizer than it would be for the user to align the edges of a circular face of the adapter with the edges of a circular face of the motion stabilizer. The chamber can have rotational symmetry such that the cutout can be placed over the end of the motion stabilizer in any orientation or at any degree of rotation around the axis of the chamber. In one embodiment, the bottom end of the adapter body can include additional physical structures that can align, guide, and fix the adapter to the motion stabilizer.

(30) In one embodiment, one or more motion stabilization components is placed in an internal sleeve, wherein the internal sleeve can be inserted into the motion stabilizer device **150**. In one embodiment, the motion stabilizer device **150** can be a hollow shell (e.g., cylindrical shell), and the internal sleeve containing motion stabilization components can be a cylindrical sleeve with a diameter that is smaller than the diameter of the motion stabilizer device **150**. The motion stabilization components can be located in the internal sleeve can be located in alignment with the positions illustrated in FIG. **3**A and FIG. **3**B.

(31) In one embodiment, the internal sleeve can form a receptacle for one or more motion stabilization components. For example, the internal sleeve can form cutouts along the length of the sleeve, the cutouts being sized for the one or more motion stabilization components. Each motion stabilization component can be placed or embedded in a cutout to be secured in the internal sleeve.

In one embodiment, the internal sleeve can be a hollow cylinder, wherein motion stabilization components is placed inside the hollow cylinder. The internal sleeve can enable placement of the motion stabilization components in the motion stabilizer device and removal of the motion stabilizer components from the motion stabilizer device as needed. In one embodiment, the cosmetic securement device can include a locking mechanism and/or an eject mechanism to secure and release the internal sleeve and the motion stabilizer components contained therein.

(32) In one embodiment, the motion stabilizer device **150** can include a strap adjustment system **405** disposed at an end of the handle portion **151**. Many users with disabilities can struggle to use both hands, as well as using a pinching motion or similar motion. Therefore, it can be difficult to adjust the strap **1103** using a pull or push motion with a single hand. Thus, the strap adjustment system **405** can be configured to adjust a fit of the strap **1103** around a hand of the user without the need of fine, dexterous manipulations. The strap **1103** can be attached or secured to the motion stabilizer device **150** at a first end of the strap **1103**, and the strap adjustment system **405** can receive a second end of the strap **1103** through one or more openings and adjust an amount of the strap **1103** pulled or pushed through the one or more openings. In doing so, a size of a loop formed between the motion stabilizer device **150** and the strap **1103** can be adjusted to be narrower or wider by the strap adjustment system.

(33) The first end of the strap **1103** can be attached or secured to, for example, the handle portion **151** or the receiver portion **152**. The location of the attachment of the first end of the strap **1103** can be modified based on a shape or structure of the motion stabilizer device **150** in order to fit a width of most user's hands for a single attachment location. Additionally or alternatively, the location of the attachment of the first end of the strap **1103** can be dynamic or adjustable in order to allow further customization according to the width of the user's hand. The width of the user's hand can be defined as, for example, a distance measuring across a palm of the user's hand in a direction perpendicular to a length axis of the arm. That is, the width of the hand can be measured from a pinkie finger to a pointer finger at the outer-most edges of said fingers. Upon grasping the motion stabilizer device **150**, the strap **1103** can be disposed along a back of the user's hand (opposite the palm side). In most cases, the user can arrange the user's thumb external to the loop in order to wrap the thumb around the handle portion **151**, but cases where the user arranges the user's thumb inside the loop can occur as well. Additional cases can occur where, for example, the user does not have a thumb or other fingers, and the strap **1103** can be adjusted accordingly for a better fit without dexterous manipulations.

(34) FIG. **5** shows an illustration of a strap adjustment system **405***a* with friction locks, according to an embodiment of the present disclosure. In an embodiment, the strap adjustment system **405***a* includes a locking spring **410**, a housing cap **415**, and a roller wheel **420**. The strap **1103** passes through a first opening and a second opening formed between the handle portion **151** and the housing cap **415**. The locking spring **410** and the roller wheel **420** are disposed between the first opening and the second opening in a chamber of the strap adjustment system **405***a*. The locking spring **410** is attached to a first side of the chamber proximal to the handle portion **151**, and the roller wheel **420** is attached to a second side of the chamber proximal to the housing cap **415**. The roller wheel **420** includes an axle around which the roller wheel **420** rotates. The axle can be secured at one or both ends to the housing cap **415**. For example, a first end of the axle and a second end of the axle are disposed in a first axle housing and a second axle housing formed as part of the second side of the chamber, the first axle housing and the second axle housing each including an elongated opening spanning along the first and second direction through which the first end of the axle and the second end of the axle can travel between the engaged and unengaged states of the roller wheel **420**. The axle, and therefore the roller wheel **420**, is configured to move along a direction towards or away from the locking spring **410** (up-down direction). The roller wheel **420** is configured to contact the strap **1103** along a first side of the strap **1103**, the first side being opposite a second side of the strap **1103** configured to contact the locking spring **410**.

(35) As shown in FIG. **5** on the left, the locking spring **410** applies a force (downwards force) against the second side of the strap **1103** towards the housing cap **415** and the roller wheel **420**, which causes the strap **1103** to deform along a portion of the strap **1103** disposed inside the chamber. For example, the locking spring **410** can be biased towards the strap **1103** via a stored spring force or a magnetic force. For example, the locking spring **410** is a magnet biased towards the strap **1103** via the magnetic force (e.g., the magnetic force is generated via a same polarity magnet pole disposed proximal to the locking spring **410** magnet pole biasing the locking spring **410** towards the strap **1103**).
(36) In an embodiment, the strap **1103**, when deformed and forced down towards the housing cap **415**, abuts a first locking edge **415***a* of the housing cap **415** at a first location of the strap **1103** and a second locking edge **415***b* of the housing cap at a second location of the strap **1103**. The abutment of the strap **1103** at the first locking edge **415***a* and the second locking edge **415***b* results in sufficient friction force to prevent translation of the strap **1103** (left or right motion). The first locking edge **415***a* and the second locking edge **415***b* are formed as part of the housing cap **415** and as part of the first opening and the second opening. Although shown as smooth, the first side and/or the second side of the strap **1103** can include ridges, bumps, serrations, valleys, etc. or other friction features to provide additional resistance to translation when the strap **1103** abuts the first locking edge **415***a* and the second locking edge **415***b*. The same friction features or complementary features can be formed on a surface of the locking spring **410** to further resist translation of the strap **1103**. The same features or complementary features can be formed on a surface of the roller wheel **420** to assist in translation of the strap **1103** via the roller wheel **420**. Further, it may be appreciated that only one of the first locking edge **415***a* or the second locking edge **415***b* can be configured to prevent the strap **1103** from translating. It may be appreciated that additional locking edges can be used to prevent the strap **1103** from translating.
(37) In an embodiment, on the right of FIG. **5**, the roller wheel **420** can be forced towards the handle portion **141** (upwards) to engage the strap **1103**, straighten the strap **1103**, and depress the locking spring **410**. In an embodiment, the upwards motion reduces the friction between the strap **1103** and the first locking edge **415** and the second locking edge **415***b* to allow translation of the strap **1103** left or right. The translation of the strap **1103** left or right is achieved by rolling the roller wheel left or right while forcing the roller wheel **420** upwards. For example, the user can grasp the motion stabilizer device **150** along the handle portion **151**, arrange the roller wheel **420** on a flat surface, such as a table, and apply a force towards the flat surface while sliding the motion stabilizer device **150** along the left or right direction. Of course, left and right directions are described, but other opposing directions can be used. For example, the user can be sitting and arrange the roller wheel **420** on a leg of the user, then push away from or pull towards the user's body. For example, the user can arrange the roller wheel **420** on a vertical surface, then move the motion stabilizer device **150** up or down.
(38) In an embodiment, the roller wheel **420** is configured to rotate around an axis of the motion stabilizer device **150** (the up-down axis as shown) such that any motion by the user will translate the strap **1103** to tighten or loosen the strap **1103** around the user's hand. For example, the roller wheel **420** can be mounted on a swiveling platform formed as part of the housing cap **415**. This can be especially helpful when the user has a reduced range of motion and can only move the motion stabilizer device **150** along a predetermined direction which is not parallel to the translation direction of the strap **1103**. Therefore, the roller wheel **420** is configured to swivel and then roll along the axis of motion of the user, and the roller wheel **420** is mated to one or more additional roller wheels, gears, and screws to convert the rolling motion of the roller wheel **420** into translation of the strap **1103**.
(39) As described above, the translation of the strap **1103** left or right via the roller wheel **420** results in a tightening or loosening of the strap **1103** around the user's hand. For example, since the strap **1103** is attached to the motion stabilizer device **150** at the first end of the strap **1103**, the

pulling of the strap **1103** at the second end of the strap **1103** through the chamber tightens the strap **1103**. Conversely, applying the opposite force or motion (pushing the second end back through the chamber) loosens the strap **1103**. Upon releasing the force biasing the roller wheel **420** towards the locking spring **410**, the locking spring **410** once again deforms the strap **1103** inside the chamber towards the roller wheel **420** and forces the strap **1103** to abut the first locking edge **415***a* and the second locking edge **415***b* to lock the strap **1103** at the current setting and tightness. Furthermore, for the strap **1103** with the first end of the strap **1103** adjustably or dynamically attached to the motion stabilizer device **150**, the user can tighten the strap **1103** to a first resistance checkpoint where the strap **1103** is taught against the back of the user's hand (e.g., by rolling the roller wheel **420**). Then, additional rolling of the roller wheel **420** results in translation of the attachment point of the first end of the strap **1103** towards the housing cap **415**. This reduces a diameter of the loop along an axis of the motion stabilizer device **150** for a more secure or snug fit with a narrow width around the user's hand.

(40) FIG. **6** shows an illustration of a strap adjustment system **405***b* with a locking pin **430**, according to an embodiment of the present disclosure. In an embodiment, the strap adjustment system **405***b* includes the locking pin **430**, a fulcrum **435**, the housing cap **415**, and the roller wheel **420**. The strap **1103** passes through the first opening and the second opening formed between the handle portion **151** and the housing cap **415**. The locking pin **430** and the roller wheel **420** are disposed between the first opening and the second opening in the chamber. Notably, the strap **1103** is not deformed or deflected in the chamber to prevent movement of the strap **1103**.

(41) In an embodiment, the strap **1103** includes teeth on the first side and the locking pin **430** has a complementary shape to the teeth. The locking ping **430** is configured to abut and mate with one of the teeth when in a locked or engaged position. The locking pin **430** can be attached to a lever arm on a first end with the roller wheel **420** attached to a second end of the lever arm. The fulcrum **435** is disposed between the locking pin **430** and the roller wheel **420**, and the lever arm is attached to the fulcrum **435**. Therefore, the lever arm pivots about the attachment point on the fulcrum **435** and causes the locking pin **430** and the roller wheel **420** to move in opposite directions to one another. As shown in FIG. **6**, when the roller wheel **420** is moved in a direction towards the strap **1103** (upwards), the locking pin **430** can move away from the strap **1103** (downwards) and disengage from the teeth. Notably, a force can be applied to bias the locking pin **430** towards the strap **1103** (upwards), such as a spring force or a magnetic force, which then constantly biases the roller wheel **420** away from the strap **1103** (downwards) via the fulcrum **430** pivot point.

(42) Again, the roller wheel **420** includes the axle around which the roller wheel **420** rotates. The axle here is attached at one or both ends to the lever arm (or lever arms when the axle is attached at both ends of the axle). The roller wheel **420** is configured to contact the strap **1103** along the first side of the strap **1103**. In an embodiment, the roller wheel **420** includes teeth as well that are complementary in shape to the teeth on the strap **1103**, thereby providing a more secure translation of the strap **1103** when the roller wheel **420** is forced upwards to contact the strap **1103** and subsequently rotate left or right to tighten or loosen the strap **1103**.

(43) In an embodiment, when the roller wheel **420** is forced to contact the strap **1103**, the locking pin **430** disengages or moves to an unlocked position due to the fulcrum, which allows the strap **1103** to move freely. At the same time, the roller wheel **420** comes into contact with the strap and, when a force is applied to rotate the roller wheel **420**, the roller wheel **420** translates the strap **1103** left or right. For example, as previously described, the user can set the motion stabilizer device **150** down on a flat surface with the roller wheel **420** end of the motion stabilizer device **150** proximal to the flat surface. Then, the user pushes down with additional force to depress the roller wheel **420** in a direction towards the strap **1103**, which releases the locking pin **430** from the strap **1103**. Then, the user moves the motion stabilizer device **150** left or right to tighten or loosen the strap **1103** around the user's hand. Upon reaching a desired fit for the strap **1103**, the user lifts the motion stabilizer device **150** off of the flat surface, which (via the spring force biasing the locking pin **430**

towards the strap **1103**) moves the locking pin **430** back into the locked or engaged position and prevents the strap **1103** from adjusting further.

(44) FIG. **7** shows an illustration of a strap adjustment system **405***c* with a ratchet button **445**, according to an embodiment of the present disclosure. In an embodiment, the strap adjustment system **405***c* includes the ratchet button **445**, a ratchet wheel **440**, and a locking pawl **450**. The strap **1103** passes through the first opening and the second opening formed between the handle portion **151** and the housing cap **415**. The ratchet wheel **440** is disposed adjacent to the strap **1103**, such as disposed along the handle portion **151**, and contacts the strap **1103** on the second side of the strap **1103** (as shown). In an embodiment, the ratchet wheel **440** is disposed along the housing cap **415** and contacts the strap **1103** on the first side of the strap **1103**. As shown, the ratchet wheel **440** pushes the strap **1103** towards the housing cap **415** to achieve sufficient friction between the ratchet wheel **440** and the strap **1103**. As described previously, the strap **1103** can optionally include the teeth that can abut and mate with complementary features formed on the ratchet wheel **440** to provide additional security in translating the strap **1103** via rotation of the ratchet wheel **440**.

(45) In an embodiment, the ratchet wheel **440** is configured to abut the strap **1103** and translate the strap **1103** laterally (towards the left or right direction) by a rotation of the ratchet wheel **440**. To this end, the ratchet button **455** is disposed on an external surface of the housing cap **415** and configured to move in a direction towards and away from the ratchet wheel **440**. To rotate the ratchet wheel **440**, the ratchet button **445** is mechanically mated to the ratchet wheel **440** and configured to be depressed to cause a rotation or ratcheting of the ratchet wheel **440**. For example, a series of gears and screws can translate the linear motion of the ratchet button **445** to the rotation of the ratchet wheel **440**. The depression of the ratchet button **445** additionally and simultaneously overcome a retention of the strap **1103** in the current position via the locking pawl **450**. The locking pawl **450** is configured to contact the ratchet wheel **440** and prevent the rotation of the ratchet wheel **440**. The locking pawl **450** can be biased towards the ratchet wheel **440** via, for example, a spring force or a magnetic force. Notably, the depression of the ratchet button **445** can result in a greater biasing force than a locking force of the locking pawl **450** on the ratchet wheel **440** to rotate the ratchet wheel **440** and thus translate the strap **1103**. It may be appreciated that the ratchet button **445** can be disposed along other external surfaces of the motion stabilizer device **150**, such as along an external surface of the handle portion **151** so the user can use a finger to depress the ratchet button **445**.

(46) In an embodiment, the locking pawl **450** is configured to contact the strap **1103** and prevent the translation of the strap **1103** (and by association, prevents the rotation of the ratchet wheel **440**). The locking pawl **450** can be biased towards the strap **1103** via, for example, a spring force or a magnetic force. Notably, the depression of the ratchet button **445** can result in a greater biasing force than a locking force of the locking pawl **450** on the strap **1103** to allow translation of translate the strap **1103**.

(47) In an embodiment, the ratchet button **445** is configured to return to a starting position via, for example, a spring or magnetic force upon reduction of the force to depress the ratchet button **445**. Thus, the ratchet button **445** can be depressed multiple times in order to rotate the ratchet wheel **440** (and release the locking pawl **450**) to translate the strap **1103**. Each time the ratchet button **445** is depressed, the ratchet wheel **440** rotates and translates the strap **1103**. Then, the locking pawl **450** prevents additional translation of the strap **1103** indirectly by preventing rotation of the ratchet wheel **440** and/or directly by preventing translation of the strap **1103**, e.g., via friction.

(48) In an embodiment, the multiple depressions of the ratchet button **445** tightens the strap **1103** around the user's hand. For example, the user can grasp the motion stabilizer device **150** along the handle portion **151**, arrange the ratchet button **445** proximal to a flat surface, such as a table, and apply a force towards the flat surface to depress the ratchet button **445** as many times as needed to translate the strap **1103** and tighten the strap **1103** around the user's hand. Upon reaching a desired fit, the user can stop depressing the ratchet button **445**.

(49) In an embodiment, the motion stabilizer device **150** can include a loosening feature, such as another button, that the user can engage to release or unlock the ratchet wheel **440** from the strap **1103** or the locking pawl **450** from the strap **1103**, thereby allowing the strap **1103** to move freely.
(50) In an embodiment, a force can be applied on the strap **1103** in the opposite direction that overcomes the locking force of the locking pawl **450** on the ratchet wheel **440**, the locking force of the locking pawl **450** on the strap **1103**, or the locking force (e.g., a friction force) between the ratchet wheel **440** and the strap **1103**. For example, the user can secure the motion stabilizer device **150** between the user's legs and pull the user's hand away from the motion stabilizer device **150** to pull the strap **1103** away from the stabilizer device **150**, which overcomes any locking force on the strap **1103** and loosens the strap **1103**.
(51) FIG. **8** shows an illustration of a strap adjustment system **405***d* with automated strap adjustment, according to an embodiment of the present disclosure. In an embodiment, the strap adjustment system **405***d* includes a motor **455**, an adjustment wheel **465** connected to the motor **455**, processing circuitry, and a button **460**. The strap **1103** passes through the first opening and the second opening formed between the handle portion **151** and the housing cap **415**. The wheel **465** is disposed adjacent to the strap **1103**, such as disposed along the handle portion **151**, and contacts the strap **1103** on the second side of the strap **1103** (as shown). In an embodiment, the wheel **465** is disposed along the housing cap **415** and contacts the strap **1103** on the first side of the strap **1103**. As shown, the wheel **465** pushes the strap **1103** towards the housing cap **415** to achieve sufficient friction between the wheel **465** and the strap **1103**. As described previously, the strap **1103** can optionally include the teeth that can abut and mate with complementary features formed on the wheel **465** to provide additional security in translating the strap **1103** via rotation of the wheel **465**.
(52) The motor **455** can be disposed in the handle portion **151**, such as at an end of the handle portion **151** proximal to the strap **1103**. The processing circuitry is electrically connected to the motor **455** and the button **460** and configured to actuate the motor **455**. The button **460** is disposed, for example, along the external surface of the housing cap **415** (as shown) or another external surface of the motion stabilizer device **150** where the user can actuate the button **460** with the user's finger.
(53) In an embodiment, the button **460** is electrically connected to the processing circuitry and configured to send an electronic signal to the processing circuitry to actuate the motor **455** upon being depressed. The motor **455**, being mechanically connected to the adjustment wheel **465**, rotates the adjustment wheel **465** to translate the strap **1103**. Again, the motor **455** (similar to the ratchet wheel **440** and the ratchet button **445**) can be mechanically connected to the adjustment wheel **465** via a set of gears and/or screws that translate a mechanical output of the motor **455** to a rotation of the adjustment wheel **465**.
(54) In an embodiment, depressing the button **460** according to predetermined patterns can result in different actuations of the motor **455**. For example, a single depression of the button **460** can cause the motor **455** to translate the strap **1103** in a first direction and a double-depression in quick succession of the button **460** can cause the motor to translate the strap **1103** in a second direction. In doing so, the single or double depression can tighten or loosen the strap. The motor **455** can be configured to run for a predetermined length of time or continuously until another action on the button **460** is performed. For example, after the double-depression of the button **460**, the motor **455** can run and rotate the adjustment wheel **465** until the button **460** is depressed again (e.g., a single depression or another double-depression), at which point the motor **455** can stop.
(55) In an embodiment, the motor **455** includes a force sensor **470** configured to determine a resistance force of the motor **455** as it acts upon the adjustment wheel **465** and thus the strap **1103**. For example, the motor **455** can be rotating the adjustment wheel **465** to tighten the strap **1103** on the user's hand and, upon the force sensor **470** detecting a resistance threshold has been reached, the force sensor **470** can stop the motor **455**. In doing so, the strap **1103** will not be tightened beyond a predetermined force and reduces the likelihood of injuring the user.

(56) In an embodiment, the PCBA **1302** is communicatively coupled to a remote processing device and configured to update the remote processing device with regards to the arrangement of the motion stabilizer device **150** and other components, such as the button **460** and its actuation. In an embodiment, the remote processing device is configured to receive data from the PCBA **1302** and transmit instructions to the PCBA **1302** based on the arrangement of the motion stabilizer device **150**. In an embodiment, the remote processing device is configured to receive input from the user regarding arrangement settings, such as a setting for the button **460** actuation and/or the resistance threshold before instructing the PCBA **1302** to adjust the strap **1103** (e.g., via the motor **455**). In an embodiment, the PCBA **1302** (via the included sensors) detects the unlocking force and other related sensor data and transmits the related sensor data to the remote processing device and awaits receiving instructions from the remote processing device based on the related sensor data. Upon receiving the instructions from the remote processing device, such as to tighten or loosen the strap via the motor **455** based on the related sensor data, the PCBA **1302** adjusts the strap **1103**.
(57) FIG. **9** shows an illustration of the strap adjustment system **405** with the adjustable or dynamic attachment of the strap **1103**, according to an embodiment of the present disclosure. In an embodiment, as previously described, the location of the attachment of the first end of the strap **1103** can be dynamic or adjustable in order to allow further customization according to the width of the user's hand. As shown, the strap **1103** is attached or fastened to an adjustable strap attachment **475** and loops around the back of a hand **499**. The user can tighten the strap **1103** to the first resistance checkpoint where the strap **1103** is taught against the back of the user's hand (e.g., by rolling the roller wheel **420**, ratcheting the strap **1103** via the ratchet button **445**, actuating the motor **455** via the button **460**, etc.). At this point, the first end of the strap **1103** at the adjustable strap attachment **475** can have remained in the same position. Then, additional efforts to tighten the strap **1103** results in translation of the adjustable strap attachment **475** (and thus the first end of the strap **1103**) towards the housing cap **415**. This reduces the diameter of the loop along an axis of the motion stabilizer device **150** for a more secure or snug fit with a narrow width around the hand **499**. For example, the strap **1103** and the adjustable strap attachment **475** can be translated to a second resistance checkpoint where additional tightening of the strap **1103** is prevented to avoid injury.
(58) In one embodiment, the present disclosure relates to a strap adjustment system for a cosmetic applicator configured for users with limited mobility, including a housing cap disposed at an end of the cosmetic applicator, the housing cap and the end of the cosmetic applicator forming a chamber including a first opening and a second opening; a strap attached to the cosmetic applicator at a first end of the strap, a second end of the strap configured to pass through the first opening and the second opening; a locking feature disposed on a first side of the chamber and configured to (i) contact a first side of the strap, (ii) bias the strap in a first direction away from the first side of the chamber via a locking force, and (iii) force a second side of the strap to abut a first locking edge and a second locking edge of the chamber formed, the first locking edge formed as part of the first opening of the chamber and second locking edge formed as part of the second opening of the chamber, the locking force preventing translation of the strap through the chamber; and a roller wheel disposed on a second side of the chamber opposite the first side of the chamber, the roller wheel configured to (i) abut the second side of the strap when an unlocking force is applied in a second direction opposite the first direction to force the roller wheel into an engaged state, the unlocking force being sufficient to overcome the locking force applied by the locking feature and allow the strap to translate through the chamber, (ii) rotate in the engaged state to translate the strap through the chamber, and (iii) return to an unengaged state where the roller wheel does not provide sufficient unlocking force to overcome the locking force applied by the locking feature on the strap.
(59) In an embodiment, the locking feature is a reversibly deformable structure configured to apply the locking force in the first direction via a spring force.
(60) In an embodiment, the locking feature is a magnet configured to apply the locking force in the

first direction via a magnetic force.
(61) In an embodiment, the strap is reversibly deformed in the chamber via the locking force applied by the locking feature, the deformation of the strap increasing a friction force between the strap at the first locking edge and the second locking edge.
(62) In an embodiment, the roller wheel is reversibly biased away from the strap in the first direction.
(63) In an embodiment, the roller wheel includes an axle disposed through a rotation axis of the roller wheel, a first end of the axle and a second end of the axle being disposed in a first axle housing and a second axle housing, respectively, formed as part of the second side of the chamber, the first axle housing and the second axle housing each including an elongated opening along the first and second direction through which the first end of the axle and the second end of the axle can travel between the engaged and unengaged states of the roller wheel.
(64) In an embodiment, the roller wheel is arranged on a swiveling platform formed as part of the second side of the chamber, the swiveling platformed configured to rotate about an axis orthogonal to a rotation axis of the roller wheel and rotate the roller wheel along a direction of an applied force applied to the roller wheel.
(65) In an embodiment, the strap includes friction features on the second side of the strap configured to provide additional resistance force against translation of the strap through the chamber.
(66) In an embodiment, the roller wheel includes complementary friction features as the friction features on the second side of the strap, the complementary friction features of the roller wheel being disposed on a surface of the roller wheel and configured to abut and mate with the friction features of the strap.
(67) In an embodiment, the roller wheel is configured to translate the strap to a first resistance checkpoint.
(68) In an embodiment, the first end of the strap is attached to the cosmetic applicator via an adjustable strap attachment configured to reversibly translate in the first direction and the second direction.
(69) In an embodiment, after reaching the first resistance check point, the roller wheel is configured to further translate the strap through the chamber and adjust an arrangement of the adjustable strap attachment along the first and second direction.
(70) In an embodiment, the roller wheel is configured to translate the strap to a second resistance checkpoint after which a tightness of the strap around an object disposed within a loop of the strap cannot be tightened further.
(71) The present disclosure additionally relates to a method of adjusting a strap in a strap adjustment system for a cosmetic applicator configured for users with limited mobility, including applying an unlocking force towards a housing cap disposed at an end of the cosmetic applicator, the housing cap and the end of the cosmetic applicator forming a chamber including a first opening and a second opening, the strap attached to the cosmetic applicator at a first end of the strap, a second end of the strap configured to pass through the first opening and the second opening, the strap adjustment system additionally including a locking feature disposed on a first side of the chamber and configured to (i) contact a first side of the strap, (ii) bias the strap in a first direction away from the first side of the chamber via a locking force, and (iii) force a second side of the strap to abut a first locking edge and a second locking edge of the chamber formed, the first locking edge formed as part of the first opening of the chamber and second locking edge formed as part of the second opening of the chamber, the locking force preventing translation of the strap through the chamber, and a roller wheel disposed on a second side of the chamber opposite the first side of the chamber, the roller wheel configured to (i) abut the second side of the strap when the unlocking force is applied in a second direction opposite the first direction to force the roller wheel into an engaged state, the unlocking force being sufficient to overcome the locking force applied by the

locking feature and allow the strap to translate through the chamber, (ii) rotate in the engaged state to translate the strap through the chamber, and (iii) return to an unengaged state where the roller wheel does not provide sufficient unlocking force to overcome the locking force applied by the locking feature on the strap; and while applying the unlocking force towards the housing cap sufficient to overcome the locking force, applying a lateral force to rotate the roller wheel and translate the strap through the chamber.
(72) Obviously, numerous modifications and variations of the present invention are possible in light of the above teachings. It is therefore to be understood that within the scope of the appended claims, the invention may be practiced otherwise than as specifically described herein.

Claims

1. A strap adjustment system for a cosmetic applicator configured for users with limited mobility, comprising: a housing cap disposed at an end of the cosmetic applicator, the housing cap and the end of the cosmetic applicator forming a chamber including a first opening and a second opening; a strap attached to the cosmetic applicator at a first end of the strap, a second end of the strap configured to pass through the first opening and the second opening; a locking feature disposed on a first side of the chamber and configured to (i) contact a first side of the strap, (ii) bias the strap in a first direction away from the first side of the chamber via a locking force, and (iii) force a second side of the strap to abut a first locking edge and a second locking edge of the chamber formed, the first locking edge formed as part of the first opening of the chamber and second locking edge formed as part of the second opening of the chamber, the locking force preventing translation of the strap through the chamber; and a roller wheel disposed on a second side of the chamber opposite the first side of the chamber, the roller wheel configured to (i) abut the second side of the strap when an unlocking force is applied in a second direction opposite the first direction to force the roller wheel into an engaged state, the unlocking force being sufficient to overcome the locking force applied by the locking feature and allow the strap to translate through the chamber, (ii) rotate in the engaged state to translate the strap through the chamber, and (iii) return to an unengaged state where the roller wheel does not provide sufficient unlocking force to overcome the locking force applied by the locking feature on the strap.
2. The system of claim 1, wherein the locking feature is a reversibly deformable structure configured to apply the locking force in the first direction via a spring force.
3. The system of claim 1, wherein the locking feature is a magnet configured to apply the locking force in the first direction via a magnetic force.
4. The system of claim 1, wherein the strap is reversibly deformed in the chamber via the locking force applied by the locking feature, the deformation of the strap increasing a friction force between the strap at the first locking edge and the second locking edge.
5. The system of claim 1, wherein the roller wheel is reversibly biased away from the strap in the first direction.
6. The system of claim 5, wherein the roller wheel includes an axle disposed through a rotation axis of the roller wheel, a first end of the axle and a second end of the axle being disposed in a first axle housing and a second axle housing, respectively, formed as part of the second side of the chamber, the first axle housing and the second axle housing each including an elongated opening along the first and second direction through which the first end of the axle and the second end of the axle can travel between the engaged and unengaged states of the roller wheel.
7. The system of claim 1, wherein the roller wheel is arranged on a swiveling platform formed as part of the second side of the chamber, the swiveling platformed configured to rotate about an axis orthogonal to a rotation axis of the roller wheel and rotate the roller wheel along a direction of an applied force applied to the roller wheel.
8. The system of claim 1, wherein the strap includes friction features on the second side of the strap

configured to provide additional resistance force against translation of the strap through the chamber.
9. The system of claim 8, wherein the roller wheel includes complementary friction features as the friction features on the second side of the strap, the complementary friction features of the roller wheel being disposed on a surface of the roller wheel and configured to abut and mate with the friction features of the strap.
10. The system of claim 1, wherein the roller wheel is configured to translate the strap to a first resistance checkpoint.
11. The system of claim 10, wherein the first end of the strap is attached to the cosmetic applicator via an adjustable strap attachment configured to reversibly translate in the first direction and the second direction.
12. The system of claim 11, wherein after reaching the first resistance check point, the roller wheel is configured to further translate the strap through the chamber and adjust an arrangement of the adjustable strap attachment along the first and second direction.
13. The system of claim 12, wherein the roller wheel is configured to translate the strap to a second resistance checkpoint after which a tightness of the strap around an object disposed within a loop of the strap cannot be tightened further.
14. A method of adjusting a strap in a strap adjustment system for a cosmetic applicator configured for users with limited mobility, comprising: applying an unlocking force towards a housing cap disposed at an end of the cosmetic applicator, the housing cap and the end of the cosmetic applicator forming a chamber including a first opening and a second opening, the strap attached to the cosmetic applicator at a first end of the strap, a second end of the strap configured to pass through the first opening and the second opening, the strap adjustment system additionally including a locking feature disposed on a first side of the chamber and configured to (i) contact a first side of the strap, (ii) bias the strap in a first direction away from the first side of the chamber via a locking force, and (iii) force a second side of the strap to abut a first locking edge and a second locking edge of the chamber formed, the first locking edge formed as part of the first opening of the chamber and second locking edge formed as part of the second opening of the chamber, the locking force preventing translation of the strap through the chamber, and a roller wheel disposed on a second side of the chamber opposite the first side of the chamber, the roller wheel configured to (i) abut the second side of the strap when the unlocking force is applied in a second direction opposite the first direction to force the roller wheel into an engaged state, the unlocking force being sufficient to overcome the locking force applied by the locking feature and allow the strap to translate through the chamber, (ii) rotate in the engaged state to translate the strap through the chamber, and (iii) return to an unengaged state where the roller wheel does not provide sufficient unlocking force to overcome the locking force applied by the locking feature on the strap; and while applying the unlocking force towards the housing cap sufficient to overcome the locking force, applying a lateral force to rotate the roller wheel and translate the strap through the chamber.
15. The method of claim 14, wherein the locking feature is a reversibly deformable structure configured to apply the locking force in the first direction via a spring force.
16. The method of claim 14, wherein the locking feature is a magnet configured to apply the locking force in the first direction via a magnetic force.
17. The method of claim 14, wherein the strap is reversibly deformed in the chamber via the locking force applied by the locking feature, the deformation of the strap increasing a friction force between the strap at the first locking edge and the second locking edge.
18. The method of claim 14, wherein the roller wheel is arranged on a swiveling platform formed as part of the second side of the chamber, the swiveling platformed configured to rotate about an axis orthogonal to a rotation axis of the roller wheel and rotate the roller wheel along the applied lateral force.

19. The method of claim 14, wherein the first end of the strap is attached to the cosmetic applicator via an adjustable strap attachment configured to reversibly translate in the first direction and the second direction.

20. The method of claim 19, wherein applying the lateral force further comprises applying the lateral force to a first resistance checkpoint where the strap tightens, to a first predetermined resistance force, around an object disposed in a loop of the strap while the adjustable strap adjustment remains in a first arrangement; and applying a second lateral force in the same direction as the lateral force to adjust the adjustable strap adjustment towards the object to further righten the strap around the object until a second resistance checkpoint is reached having a second predetermined resistance force that arranges the adjustable strap adjustment in a second arrangement.